4. COMM Protocols

'Mean of noise is small ~ 0'

24/03/25 9:14 PM

Types : [CAN] SPI I2C

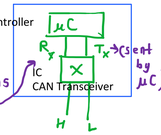
AMS (accumulator mngmnt sys t)

CCD (Controls com. & Data acquisition) → core of all computat^n

Single CAN channel has H & L each

Teensy ?

2 specified pins for CAN



We don't use one single line for transferring and receiving cuz noise aa sakta hai

$x + \epsilon$

$y + \epsilon$

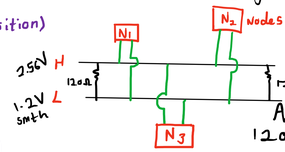
o/p → x−y cuz l of wires is equal ↳ shall stay the same

DATA Loss bhi ho sakta hai

CAN is a form of serial communicat^n.

Works on multi-master system

CAN → master system



HAS to be 120Ω.

why?

When we deal with High speed signals it takes finite amt of time to get to signal to be sent.

Signal reflects back after hitting load. & input mein reflected + sent signal ka Σ hota hai.

Line ka characteristic impea-dance is 60Ω. If we terminate wire with load thats equal to characteristic impedance, reflection nhi hota.

2× 120Ω Rs in || are $R_{eq} = 60\Omega$

Khudse read kylena.

Baud rate of all nodes must be same

e^- could get accumulated on surface

$N_1 - M_1$ → signal/message

$N_2 - M_2$, $N_3 \rightarrow M_3$

message frame

	DLC	Data	CRC

→ if this fails, koi signal nhi jayega.

Message ID aka Arbitrat^n ID → Determines priority

lowest value Vala ID will go first

[IF some ID toh ERROR]

11 bits ka hai in 2.0

length of data

DLC = No. of bytes in data.

8 bytes max

4 bytes = Nibble

(ENDIAN-ness in assignment)

Min^m 2 Nodes are req^d for it to work.

At any time kisi bhi message ID ko access kr skte hai, DATA in it can change.

Baud Rate: Bits/s or frames/s

Max speed with which my data can be transferred.

8 msg → 125 bits/ms

every 10 millisec. I transmit 1 msg to bus

all 8 are called CALL

$(8 \times 100 \times 125 \text{ bits/s})$ → $\frac{1000}{10}$ ms

Bus Load = $\frac{\text{Actual Speed}}{\text{Baud rate}} \times 100$

thumb rule : this should be < 30%

> 30% toh after some time CAN fails

8 msg in BUS at t^t

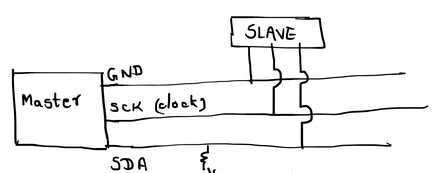
unka value might not change.

We 'read' any 1 if we want.

SPC ll read hote hai

Serial to parallel converter

I2C



Master can initiate communicat^n

7 or 10 bit

	ID	Send 8 bytes	

Start Bit

which slave → if 0 at end M wants to WRITE

CLK ↑
SDA ↓ } all slaves are aware that master is ready to transmit

1 → acknowledgement

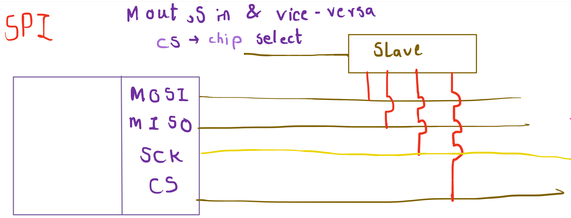
after this M waits for acknowledgement from slave.

If SDA low, slave is ready to receive

SPI

M out, S in & vice-versa

CS → chip select

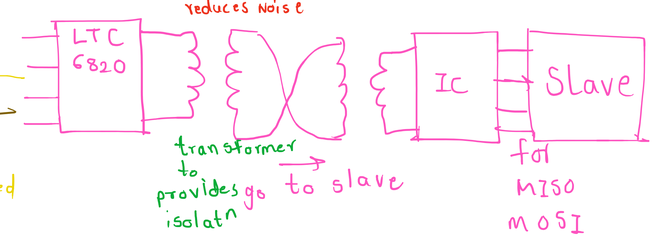


CS_2
CS_3
CS_4 ..... tell which slave is to be selected

AMS uses iso-SPI (it reduces noise)

converting 4 wires to 2 wires

Reduces Noise



Daisy Chain configurat^n